It is a classical optimization problem.

It is the problem of finding the most valuable combination of items to include in a knapsack (container) with a limited weight capacity.

Given a set of item, each with a weight and profit profit value, determine the combination of items to include in the knapsack so capacity of the knapsack and the total value of the knapscale and the value of the item should be maximized

Let n be the no of objects, each objects is having a weight weight and contribution to profit.

Let m be the knapsack capacity.

The objective is to feel the knapsack in such a way that the profit should be maximized

It is also allowed that a fraction of item to be added to the knapsack.

At each step we include the object which is the maximum.

Mathematically it is written as maximum $\sum p_i x_i$

$$\max \sum_{i=1} p_i x_i$$